

3.2) $V :=$ spojité reálné funkce s definičním oborem $(1,5)$ a sk. s. $\langle \cdot, \cdot \rangle$

$$\langle f, g \rangle = \int_1^5 f(x)g(x) dx$$

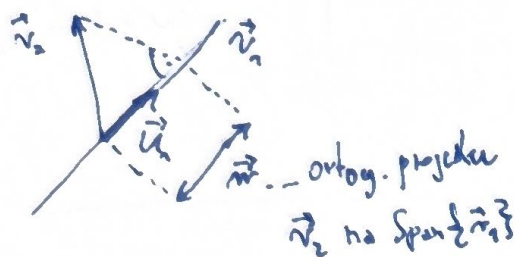
a) ON báze podprostoru $\text{Span}\{1, x, x^2\}$ prostoru V podle B-S ortogonalizace:

VSTUP: $(\vec{v}_1, \vec{v}_2, \vec{v}_3) \subset V$

VIŠTUP: $(\vec{u}_1, \vec{u}_2, \vec{u}_3)$ ON báze

$$1) \vec{u}_1 = \frac{1}{\|\vec{v}_1\|} = \frac{1}{\sqrt{5}} \quad \begin{array}{|c|} \hline 1 \\ \hline 1 \quad 5 \\ \hline \end{array}$$

$\vec{v}_2 - \vec{w}_1 \dots$ kolmice

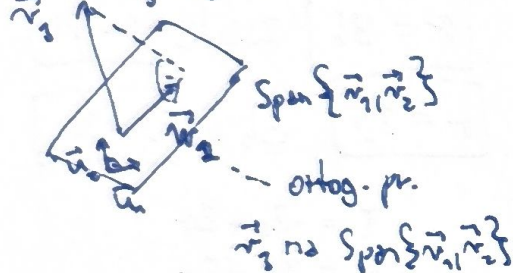


$$\vec{w}_1 = \langle \vec{v}_2, \vec{u}_1 \rangle \vec{u}_1 = \int_1^5 x \cdot \frac{1}{\sqrt{5}} dx \cdot \frac{1}{\sqrt{5}} = \frac{12}{\sqrt{5}} \cdot \frac{1}{\sqrt{5}} = \frac{12}{5}$$

$$2) \vec{u}_2 = \frac{1}{\|\vec{v}_2 - \vec{w}_1\|} (\vec{v}_2 - \vec{w}_1) = \frac{75}{508} \left(x - \frac{12}{5} \right)$$

$$\sqrt{\int_1^5 \left(x - \frac{12}{5} \right)^2 dx} = \frac{508}{75}$$

$\vec{v}_3 - \vec{w}_2 \dots$ kolmice



$$3) \vec{w}_2 = \langle \vec{v}_3, \vec{u}_1 \rangle \vec{u}_1 + \langle \vec{v}_3, \vec{u}_2 \rangle \vec{u}_2 = \frac{124}{15} + \frac{2745}{127} \cdot \frac{75}{508} \left(x - \frac{12}{5} \right)$$

$$\int_1^5 \frac{x^2}{\sqrt{5}} dx = \frac{124}{3\sqrt{5}}$$

$$\int_1^5 \frac{75x^3}{508} - \frac{12 \cdot 75}{5 \cdot 508} dx = \frac{2745}{127}$$

$$\vec{u}_3 = \frac{1}{\|\vec{v}_3 - \vec{w}_2\|} (\vec{v}_3 - \vec{w}_2) = \frac{1000}{116916} \left(x^2 - \frac{124}{15} - \frac{205875}{64516} \left(x - \frac{12}{5} \right) \right)$$

ON báze podprostoru $\text{Span}\{1, x, x^2\}$

$$\text{je } B = \left(\frac{1}{\sqrt{5}}, \frac{75}{508} \left(x - \frac{12}{5} \right), \frac{1000}{116916} \left(x^2 - \frac{124}{15} - \frac{205875}{64516} \left(x - \frac{12}{5} \right) \right) \right)$$

$$b) [\sin x]_B = \begin{pmatrix} \int_1^5 \sin x \cdot \vec{u}_1 dx \\ \int_1^5 \sin x \cdot \vec{u}_2 dx \\ \int_1^5 \sin x \cdot \vec{u}_3 dx \end{pmatrix} \approx \begin{pmatrix} 0,11477 \\ -0,48637 \\ -0,085068 \end{pmatrix}$$

$$\sin x \approx 0,11477 - 0,48637x - 0,085068x^2$$

$$c) \quad \| \sin x - (ax^2 + bx + c) \| =$$

$$= \sqrt{\int_0^{\frac{\pi}{2}} \left(\sin x - (-0,085\,068x^2 - 0,486\,37x + 0,114\,77) \right)^2 dx} \approx 4,59866$$